

Time Series Forecasting & Model Evaluation

Graduate Quantitative Economics Project | Real Personal Consumption Expenditures Per Capita: Nondurable Goods

Tool EViews	Data FRED A796RX0Q048SBEA	Period 1947Q2-2020Q1	Models AR(1)-AR(4)
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Project objective

Compare the out-of-sample forecasting performance of four autoregressive specifications for the annualized quarterly growth rate of real personal consumption expenditures per capita on nondurable goods. The analysis determined which specification produced the most reliable forecasts while testing whether model assumptions and forecast errors supported that conclusion.

Analytical approach

1	Transform	Converted the level series to annualized quarterly log growth.
2	Model	Estimated AR(1), AR(2), AR(3), and AR(4).
3	Backtest	Used an expanding-window pseudo out-of-sample design beginning with a 1947Q2-1989Q4 estimation window.
4	Evaluate	Compared 1-, 2-, 3-, and 4-quarter-ahead forecasts using MSE, median squared error, and MAE.
5	Validate	Tested residual behavior, forecast bias, calibration, error dependence, and forecast encompassing.

Key result: AR(3) produced the lowest MSE, median squared error, and MAE at every forecast horizon tested.

Why this project matters: It used repeated out-of-sample forecasting and multiple diagnostic tests to separate predictive performance from in-sample fit and identify limitations in the final recommendation.

Methodology: Testing Models as New Data Arrives

Each model was estimated using only the information available through a given quarter, forecasts were generated, and the estimation window was then expanded as new observations became available. This mimics a real-time forecasting environment and avoids evaluating forecasts with future information.

Initial estimation 1947Q2-1989Q4	Forecast 1-4 quarters ahead	Add new data Expand one quarter	Repeat Through 2020Q1
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Model comparison framework

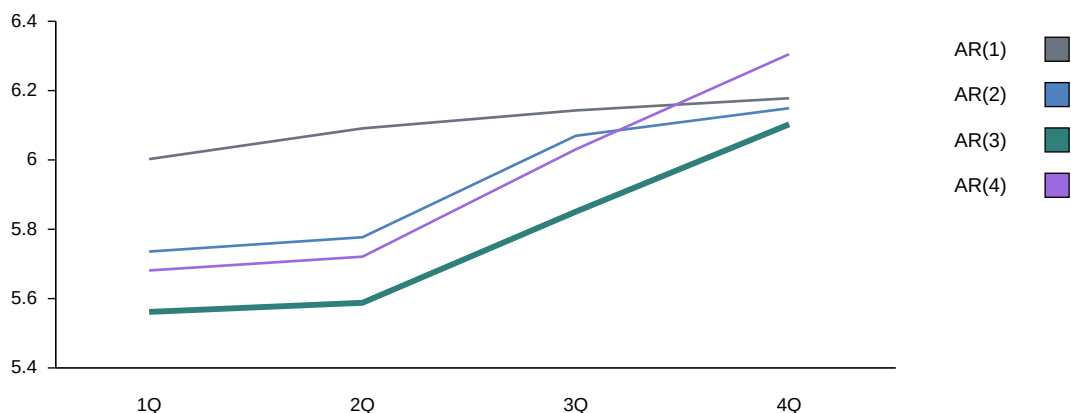
Question	Measures / Tests	What it addressed
How well does the model fit?	AIC, SIC	Fit balanced against model complexity
Are residuals behaving appropriately?	Ljung-Box, Jarque-Bera	Serial correlation and normality assumptions
How accurate are future predictions?	MSE, MEdSE, MAE	Out-of-sample forecast error
Are forecasts systematically wrong?	Bias tests	Systematic over- or underprediction
Are forecasts calibrated?	Mincer-Zarnowitz	Relationship between forecasts and realizations
Do errors retain predictable structure?	Forecast-error tests	Remaining dependence in forecast errors
Does another model add information?	Encompassing tests	Incremental predictive information

What the diagnostics showed

The in-sample criteria did not identify one universally preferred specification: average AIC slightly favored AR(2), final-window AIC favored AR(3), and SIC favored AR(1). Ljung-Box results indicated remaining autocorrelation in some windows, while Jarque-Bera tests strongly rejected residual normality. These findings made out-of-sample performance especially important.

Out-of-Sample Performance

AR(3) delivered the strongest forecast accuracy across every horizon and all three reported loss measures. Its advantage was largest at shorter horizons and narrowed at the four-quarter horizon, but the ranking remained consistent.



Mean squared error by forecast horizon (lower is better).

Horizon	AR(1) MSE	AR(2) MSE	AR(3) MSE	AR(4) MSE	Best
1 quarter	6.0024	5.7357	5.5616	5.6812	AR(3)
2 quarters	6.0910	5.7770	5.5879	5.7209	AR(3)
3 quarters	6.1428	6.0696	5.8507	6.0305	AR(3)
4 quarters	6.1779	6.1492	6.1027	6.3051	AR(3)

Interpreting the result

AR(3) also had the lowest median squared error and mean absolute error at all four horizons. Bias tests provided little evidence of systematic over- or underprediction, and the AR(3) Mincer-Zarnowitz null was not rejected at the 5% level at any horizon.

Important limitation: AR(3) did not dominate every diagnostic. Forecast-error dependence remained at some horizons, residual normality was rejected, and encompassing tests showed that competing specifications could still contain useful information in selected comparisons.

Recommendation, Limitations & Transferable Skills

Recommendation

Select AR(3) as the preferred specification among the four models tested.

It achieved the lowest MSE, median squared error, and MAE at every 1- through 4-quarter forecast horizon. The result should still be used with appropriate caution because diagnostics identified non-normal residuals and some remaining forecast-error dependence.

Analytical judgment demonstrated

Forecasting	Built and compared multiple autoregressive specifications across several forecast horizons.
Backtesting	Used an expanding-window pseudo out-of-sample design to evaluate performance on unseen periods.
Model evaluation	Compared models using multiple error measures rather than relying on a single statistic.
Quality control	Used residual and forecast diagnostics to challenge assumptions and identify limitations.
Communication	Translated statistical evidence into a model recommendation while clearly stating uncertainty.

What I would carry into a business setting

The central lesson from this project is that a model should be judged by how well it performs on the decision it is meant to support, not simply by how sophisticated it appears or how well it fits historical data. The best-performing specification was not the most complex one, and diagnostic testing showed why even a stronger forecast should be communicated with its limitations.

That same approach applies to business analytics: define the question, validate the underlying information, compare alternatives using appropriate measures, investigate unexpected results, and communicate a conclusion that is both actionable and supported by the evidence.

Project provenance

Completed as graduate coursework in Quantitative Economics/Econometrics. Analysis performed in EViews. Source: FRED, Real Personal Consumption Expenditures Per Capita: Goods: Nondurable Goods (A796RX0Q048SBEA). This work sample condenses the original 32-page research paper; no new model results were introduced.